PROJECT CASE STUDY

Texture Statistics vs Diffusion Features

Heeger-Bergen, Wavelets & U-Net Analysis

Category	Status	Year
AI Research	Research implementation	2026

A research implementation comparing fixed multiscale texture statistics with learned diffusion U-Net representations and tracking statistical convergence during denoising.

Multi-scale	Wasserstein	Hybrid
filter analysis	distance metric	synthesis study

Context and problem

Diffusion models generate convincing textures, but it is unclear how their learned features relate to classical multiscale filter banks and when generated images recover the statistics of the original texture.

Strategy and implementation

- Implemented steerable-pyramid decomposition and subband statistics.
- Compared learned U-Net filters with classical multiscale filters in the frequency domain.
- Tracked Wasserstein distances across denoising stages.
- Tested a hybrid workflow where a classical statistical draft is refined by diffusion.

My role

Research design, Python implementation, statistical analysis, visualization, tests, and interpretation of expected-vs-observed behavior.

Technology stack

Python | PyTorch | Diffusion U-Net | Steerable Pyramid | Wavelets | Wasserstein Distance

Results and evidence

- Produced filter-comparison, statistical-evolution, and hybrid-synthesis outputs.
- Built a reproducible multi-phase codebase with tests.
- Connected classical texture theory to modern generative-model representations.

Key takeaways

- Classical statistics and learned representations answer different questions.
- A strong research result reports where the expected coarse-to-fine pattern holds and where it does not.

Portfolio: github.com/awai1s | awaisraza5424@gmail.com